

ATKA, Russian Federation

WMO: 259020

Lat: 60.850N

Lon: 151.767E

Elev: 2956

StdP: 13.19

Time Zone: 11.00 (E11)

Period: 86-95

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-42.2	-38.5	-45.6	0.4	-41.7	-42.3	0.5	-38.3	27.4	-20.2	24.8	-23.3	0.6	0	0.886

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	18.3	74.7	57.4	71.1	55.5	67.4	53.6	58.5	71.8	56.5	69.1	54.5	65.7	7.5	190

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
52.5	65.4	63.3	50.3	60.3	61.3	48.5	56.2	58.7	26.8	71.9	25.3	69.2	24.1	65.7	66.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 21.7	19.4	17.6	-45.1	80.4	5.4	3.0	-49.0	82.5	-52.1	84.3	-55.2	85.9	-59.1	88.1		
(5)			-45.0	61.4	5.0	2.8	-48.6	63.5	-51.5	65.1	-54.3	66.7	-58.0	68.8		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	14.6	-22.1	-16.1	-5.4	11.1	30.7	49.2	54.5	48.1	35.3	14.3	-7.3	-19.5		
	DBStd	28.92	9.05	9.65	11.32	9.16	7.40	8.16	6.22	7.49	7.98	11.40	12.11	12.04		
	HDD50	13257	2234	1851	1717	1166	598	115	28	123	443	1107	1719	2156		
	HDD65	18411	2699	2271	2182	1616	1063	476	328	522	892	1572	2169	2621		
	CDD50	326	0	0	0	0	0	91	168	66	1	0	0	0		
	CDD65	5	0	0	0	0	0	2	3	0	0	0	0	0		
	CDH74	111	0	0	0	0	0	33	58	20	0	0	0	0		
	CDH80	5	0	0	0	0	0	1	4	0	0	0	0	0		
	WSAvg	6.8	8.0	7.6	6.5	7.7	6.8	6.8	6.1	6.6	5.9	5.8	6.9	6.4		
	PrecAvg	20.4	0.7	0.8	1.1	1.0	1.2	2.0	2.6	3.3	2.4	2.2	1.9	1.2		
(15) Precipitation	PrecMax	27.4	1.6	2.2	2.9	3.9	3.1	3.5	4.8	6.4	4.7	3.7	4.6	4.0		
	PrecMin	13.0	0.1	0.0	0.0	0.0	0.0	0.5	0.9	1.2	0.2	0.3	0.3	0.2		
	PrecStd	3.0	0.4	0.5	0.7	0.8	0.6	0.8	0.9	1.4	1.1	0.8	1.0	0.8		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	17.2	12.6	24.5	36.0	56.7	78.1	78.7	77.0	61.3	40.8	28.5	13.2		
		MCWB	15.1	11.1	22.1	29.2	45.2	58.8	60.2	58.3	47.7	35.1	25.4	11.9		
	2%	DB	0.2	5.9	18.4	31.3	51.1	73.7	75.2	71.2	54.6	37.3	23.4	7.3		
		MCWB	-0.5	4.7	15.9	27.2	41.5	56.4	58.5	55.7	43.9	33.3	21.7	6.3		
	5%	DB	-6.1	1.1	15.2	27.9	46.3	70.2	71.8	66.4	50.6	34.0	17.5	0.7		
		MCWB	-6.7	0.3	13.3	25.1	39.1	54.0	56.5	53.4	42.5	31.5	16.3	0.1		
	10%	DB	-9.4	-2.7	10.8	25.0	42.5	65.9	68.1	61.8	47.0	30.1	10.4	-4.9		
		MCWB	-9.9	-3.3	9.0	22.2	35.9	52.1	54.6	50.9	41.4	27.7	9.4	-5.6		
	0.4%	WB	14.7	11.1	21.9	29.7	45.7	59.7	62.0	58.5	48.3	37.0	25.6	12.3		
		MCDB	16.9	12.4	24.4	33.6	56.4	75.9	75.1	73.0	58.1	39.8	28.2	13.7		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	-0.2	4.8	16.0	27.6	42.0	57.1	59.5	56.3	45.4	33.8	21.7	6.5		
		MCDB	0.2	5.7	18.0	31.2	50.0	73.0	72.3	68.9	52.2	36.2	23.6	7.4		
	5%	WB	-6.8	0.4	13.3	25.3	39.1	54.8	57.6	54.3	43.6	30.8	16.1	0.1		
		MCDB	-6.3	1.1	15.2	28.3	45.7	68.2	69.1	64.6	49.2	34.0	17.2	0.6		
	10%	WB	-10.0	-3.4	9.2	22.5	36.6	52.5	55.6	51.8	41.9	27.8	9.4	-5.3		
		MCDB	-9.5	-2.7	10.9	25.0	41.5	64.9	66.9	60.4	46.7	30.1	10.3	-4.6		
	5% DB	MDBR	8.8	10.7	15.1	15.1	14.2	19.3	18.3	17.5	13.9	13.1	11.5	9.0		
		MCDBR	12.4	13.6	18.3	15.2	19.3	27.2	25.9	28.0	20.7	15.3	17.5	12.9		
		MCWBR	12.3	12.8	15.9	11.8	11.8	13.2	11.3	13.8	11.9	11.5	16.0	12.1		
		MCWBR	12.8	13.3	18.1	13.6	17.5	25.7	23.6	24.7	18.2	13.7	17.2	13.0		
(35) Mean Daily Temperature Range	5% WB	MCWBR	12.4	12.5	15.7	10.6	11.0	12.9	10.9	12.9	10.5	10.6	15.9	12.2		
(40) Clear-Sky Solar Irradiance	taub		0.191	0.237	0.285	0.357	0.384	0.369	0.417	0.385	0.302	0.262	0.204	0.165		
	taud		2.078	2.164	2.075	1.899	1.989	2.325	2.205	2.285	2.476	2.245	2.047	1.939		
	Ebn at Noon		190	244	263	259	263	272	253	252	262	228	172	141		
	Edh at Noon		12	22	34	50	49	36	40	34	23	20	12	8		
(44) All-Sky Solar Radiation	RadAvg		123	416	990	1647	1821	1653	1572	1142	768	436	171	61		
	RadStd		12	27	48	75	161	183	160	130	122	62	17	4		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (3 neighbors)	N/A	N/A	-2.90	+1.82	+1.17	+1.03	N/A	N/A	+59	N/A

Nomenclature: See separate page